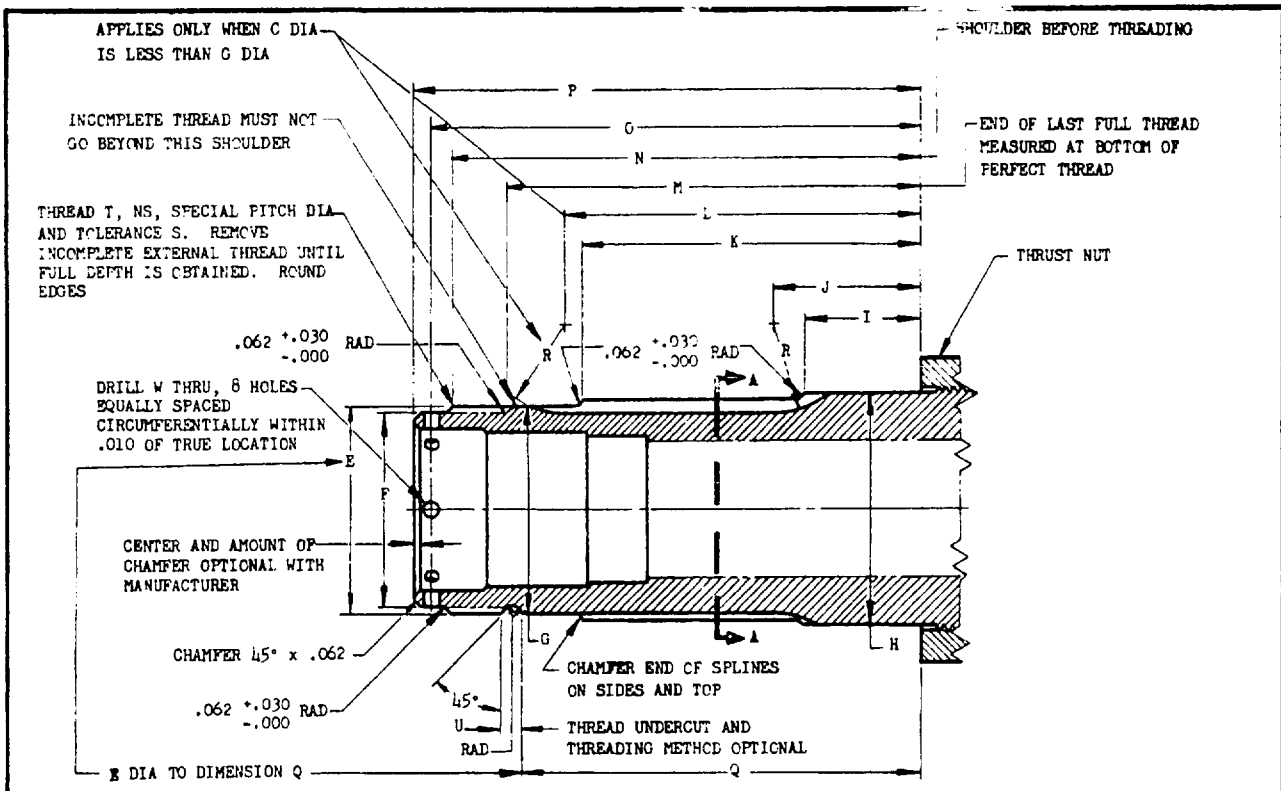
NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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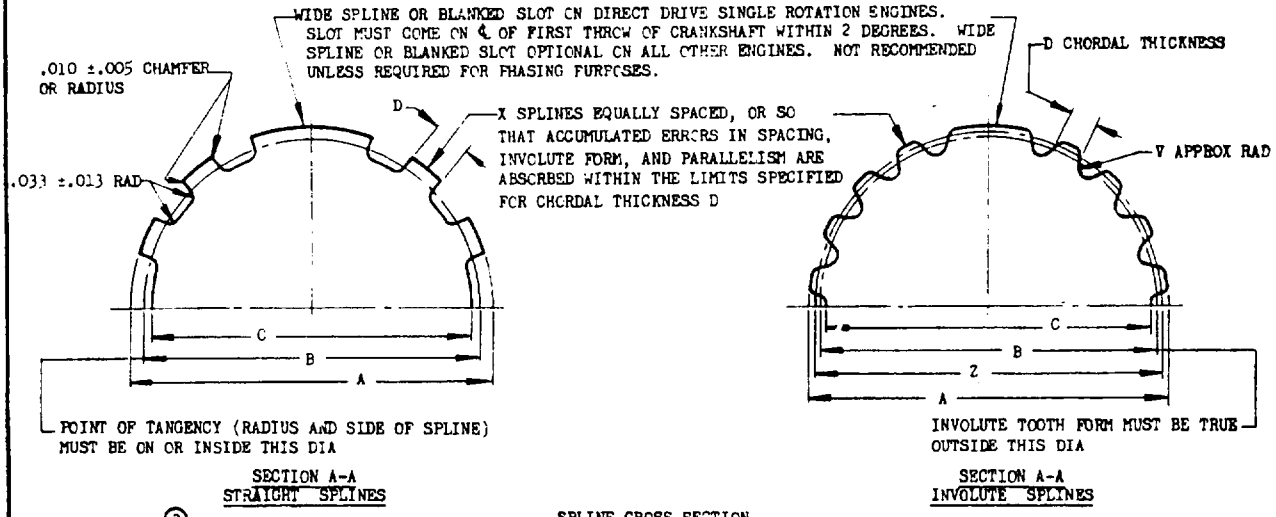


TAPER: SURFACES AT DIA G AND H NOT TO TAPER MORE THAN .0005 PER INCH OF LENGTH.

CONCENTRICITY: WITH THE SHAFT MOUNTED ON DIA G AND REAR BEARING SURFACE, DIA H, OTHER BEARING SURFACES AND DIA A ON STRAIGHT SPLINES SHALL BE CONCENTRIC WITHIN .001 FULL INDICATOR READING, THREAD T PITCH DIA S SHALL BE CONCENTRIC WITHIN .005 FULL INDICATOR READING AND DIAMETER Z ON INVOLUTE SPLINE SHALL BE CONCENTRIC WITHIN .002" FULL INDICATOR READING.

SEE TABLE I SHEET 10 FOR VALUES OF DIMENSIONS.

SINGLE ROTATION SHAFT-LONGITUDINAL SECTION



③

SPLINE CROSS SECTION

FOR VALUES OF DIMENSIONS SEE: TABLE I, SHEET 10 FOR SINGLE ROTATION SHAFTS.
TABLE II, SHEET 12 FOR DUAL ROTATION SHAFTS.

SHAFT SPLINE DATA:
X = NUMBER OF TEETH
Y = DIAMETRAL PITCH
Z = PITCH DIA (THEOR)
30° PRESSURE ANGLE

BREAK EDGES .016.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: THREE PLACE DECIMALS ±.010, ANGLES ±2°.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATIONS.

SHEETS NO. 5, 6, AND 9 ADDED.

APPROVED 21 Feb 47 REVISED (1) 24 Apr 51 (2) 24 Oct 52 (3) 10 Jun 53

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

SHAFT ENDS - PROPELLER

AND10152

SHEET 1 OF 13

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NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all antecedent standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new design, not later than 6 months after the latest date of approval shown.

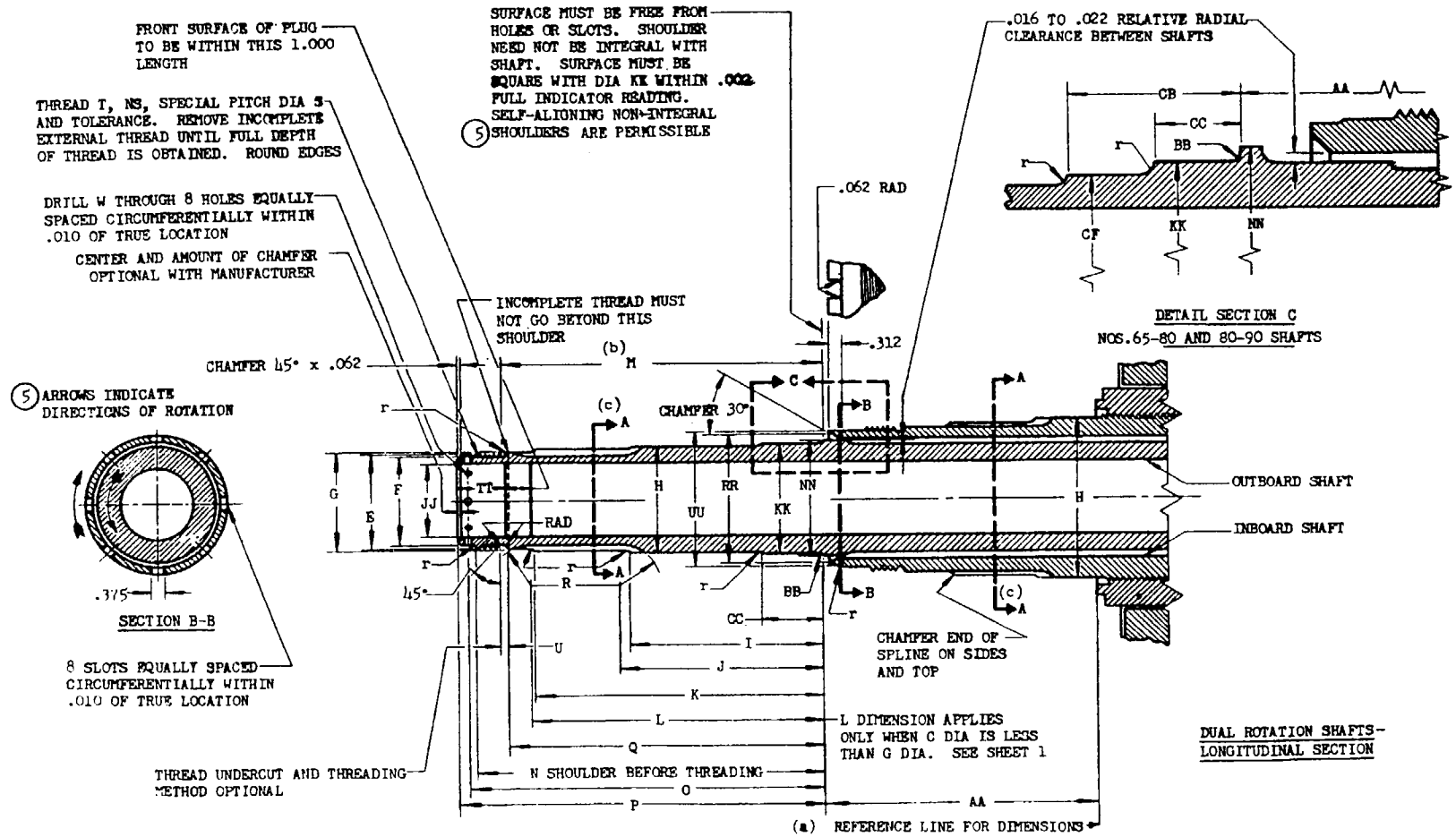
NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

SHAFT ENDS — PROPELLER

AND10152

SHEET 2 OF 13



(a) DIMENSIONS, DIMENSION LETTERS, AND NOTES ARE FOR BOTH SHAFTS AS APPLICABLE.
(b) M DIMENSION TO END OF LAST FULL THREAD MEASURED AT BOTTOM OF PERFECT THREAD.
(c) SEE SHEET 1 FOR SECTION A-A ON SPLINES.

CONCENTRICITY: WITH EITHER SHAFT MOUNTED ON DIA G AND REAR BEARING SURFACE, DIA H, KK AND CF, AS APPLICABLE, OTHER BEARING SURFACES, AND DIA A ON STRAIGHT SPLINES SHALL BE CONCENTRIC WITHIN .001 FULL INDICATOR READING. THREAD T, PITCH DIA S SHALL BE CONCENTRIC WITHIN .005 FULL INDICATOR READING; AND DIA Z ON INVOLUTE SPLINES SHALL BE CONCENTRIC WITHIN .002 FULL INDICATOR READING. DIA H ON INBOARD SHAFT AND DIA HH (SEE SHEET 4) SHALL BE CONCENTRIC WITHIN .007 FULL INDICATOR READING. WITH SHAFTS UNSUPPORTED BY PROPELLER BEARING, DIA KK ON OUTBOARD SHAFT AND DIA G ON INBOARD SHAFT SHALL BE CONCENTRIC WITHIN .018 FULL INDICATOR READING.

TAPER: SURFACE AT DIA G AND H NOT TO TAPER MORE THAN .0005 PER INCH OF LENGTH.

THE PROPELLER MANUFACTURER SHALL PROVIDE A BEARING TO FIT DIA KK OR DIA CF.

BREAK EDGES..016.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: THREE PLACE DECIMALS ±.010, ANGLES ±2°.

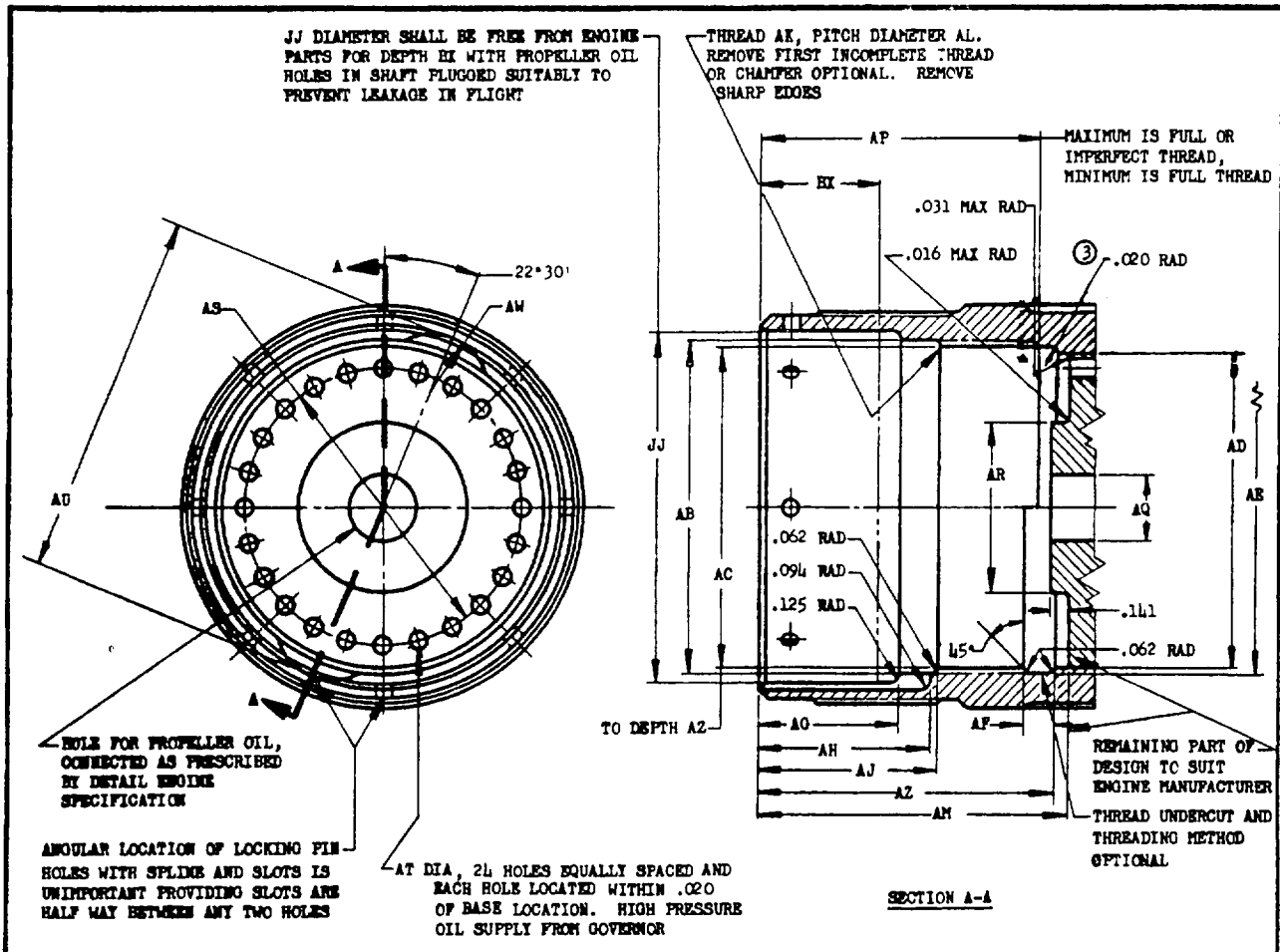
SEE TABLE II SHEETS 12 AND 13 FOR VALUES OF DIMENSIONS.

GEAR RATIOS FOR EACH SHAFT SHALL BE IDENTICAL

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

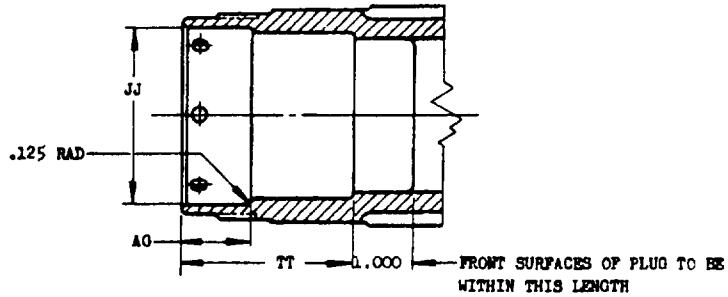
APPROVED 21 Feb 47 REVISED (1) 25 Aug 48 (2) 2 Mar 49 (3) 24 Apr 51 (4) 24 Oct 10 Jun 53

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SECTION A-A

SINGLE ROTATION SHAFT END - INTERNAL DETAILS, WHEN DESIGNED FOR HYDRAULIC-CONTROLLABLE PROPELLERS



SINGLE ROTATION SHAFT END - INTERNAL DETAILS, WHEN PLUGGED

CONCENTRICITY: DIAMETER AD, AR AND AL MUST BE CONCENTRIC WITH DIAMETER G ON SHEET 1 WITHIN .005 FULL INDICATOR READING.

BREAK EDGES .016.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: THREE PLACE DECIMALS, ±.010, ANGLES ±2°.

SEE TABLE I SHEET 11 FOR VALUES OF DIMENSIONS.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

APPROVED 21 Feb 47 REVISED (1) 24 Apr 51 (2) 24 Oct 52 (3) 10 Jun 53

This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all associated standard drawings for the same product and shall become effective for the procurement of commercial supplies, or for use in new designs, not later than 6 months after the issued date of approved sheets.

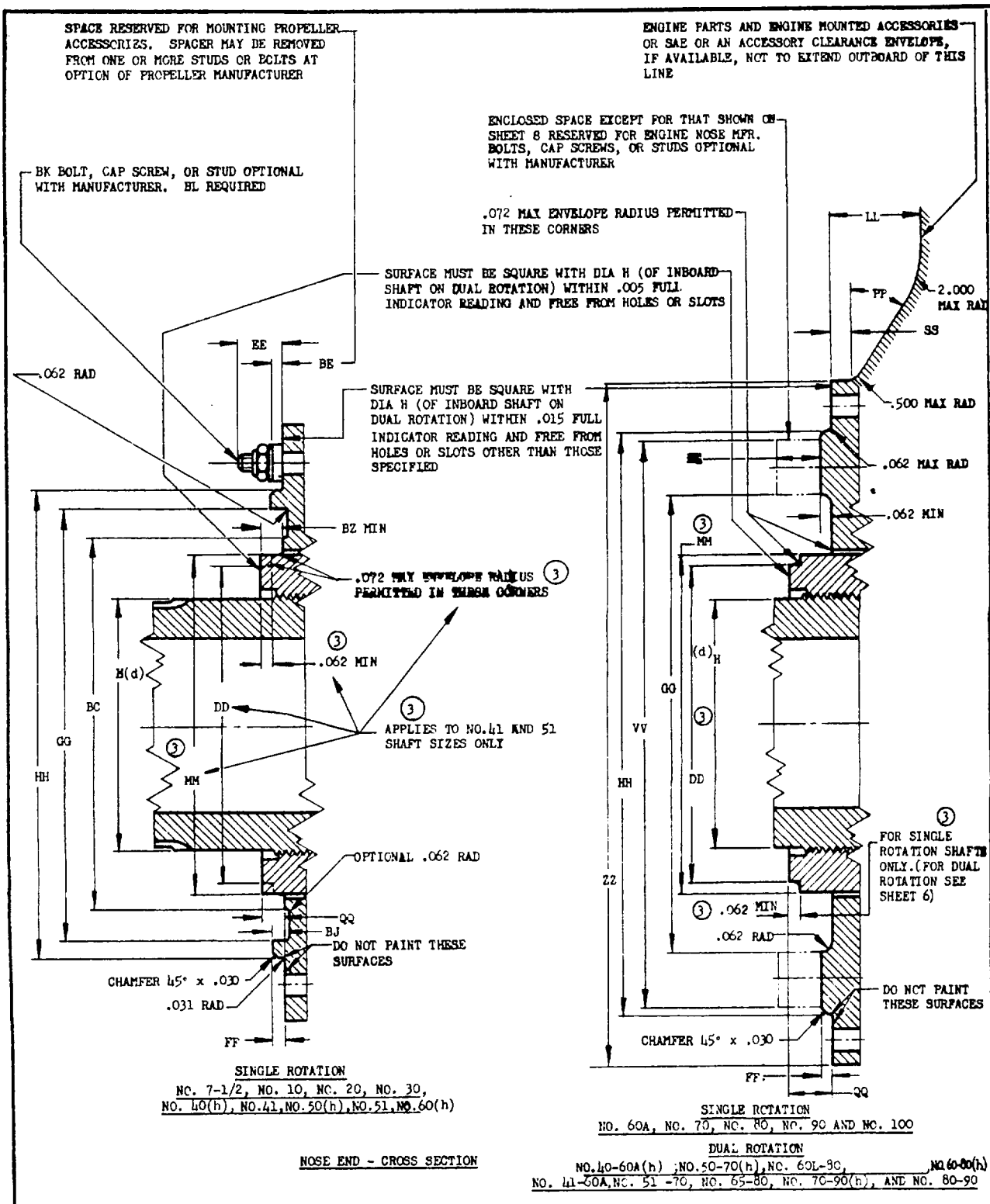
Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SHAFT ENDS - PROPELLER

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(d) FOR DIMENSION H SEE SHEET 1 OR SHEET 2, AS APPLICABLE.

(h) INACTIVE FOR DESIGN AFTER 24 APRIL 1951

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

CONCENTRICITY: DIA H AND HH TO BE CONCENTRIC WITHIN .007 FULL INDICATOR READING.

BREAK EDGES .016.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: THREE PLACE DECIMALS ±.010, ANGLES ±2°.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

SEE SHEETS 6,7 AND 8 FOR OTHER APPLICABLE DIMENSIONS AND SEE TABLES I AND II , SHEETS 10,11,12 AND 13 FOR VALUES OF DIMENSIONS.

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

SHAFT ENDS - PROPELLER

AND10152

SHEET 4 OF 13

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

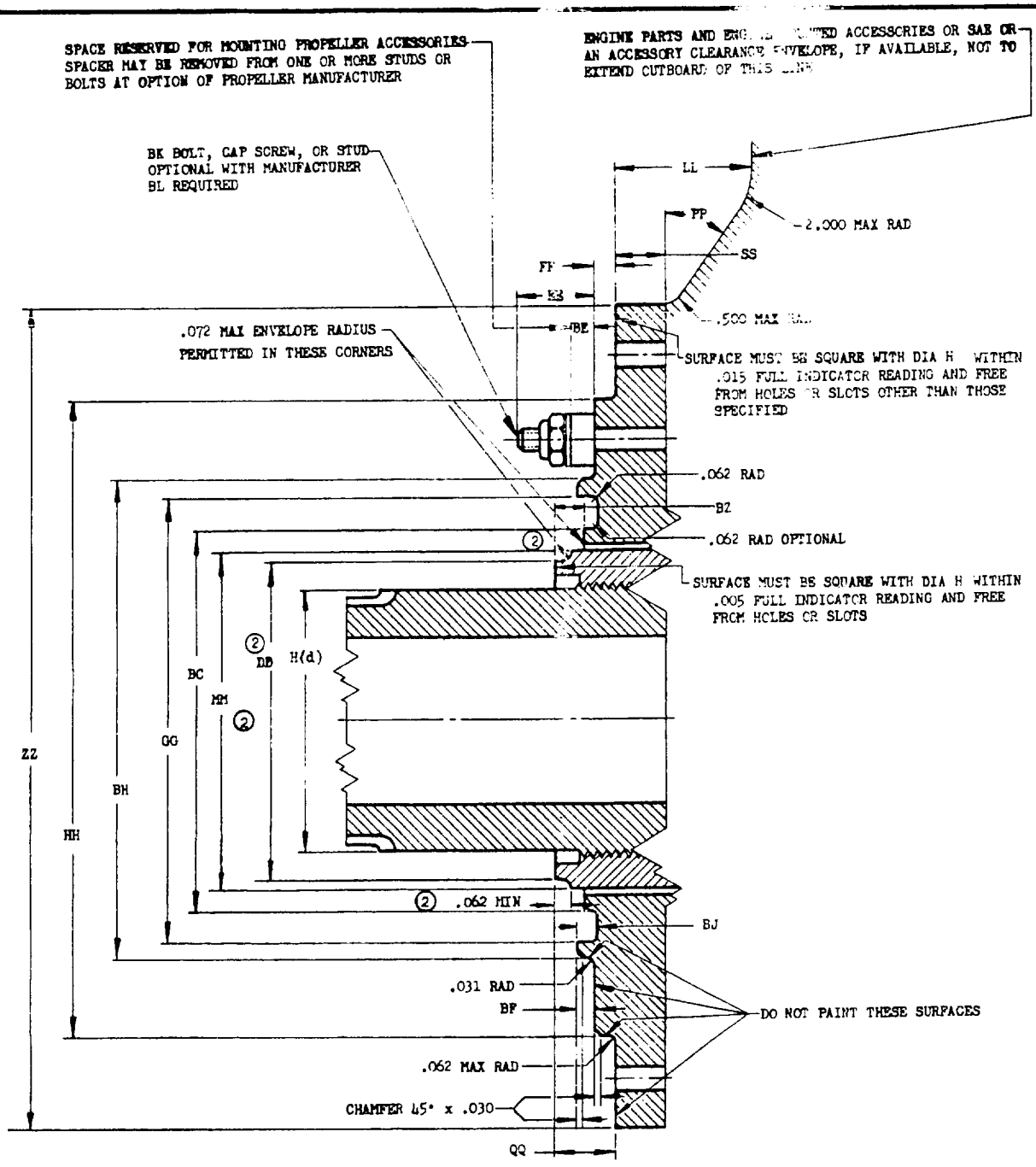
NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all antecedent standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new design, not later than 6 months after the latest date of approval shown.

REVISED 21 Feb 47 (1) 24 Apr. 51 (2) 24 Oct 52 (3) 10 Jun 53

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SINGLE ROTATION
NO. 50A(h) AND 51A

NOSE END - CROSS SECTION

(d) FOR DIMENSION H SEE SHEET 1.

(h) INACTIVE FOR DESIGN AFTER 24 APRIL 1951.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

CONCENTRICITY: DIAMETER H AND HH TO BE CONCENTRIC WITHIN .007 FULL INDICATOR READING.

(2) SEE SHEET 9 FOR OTHER APPLICABLE DIMENSIONS AND SEE TABLE I SHEETS 10 AND 11 FOR VALUE OF DIMENSIONS.

BREAK EDGES .016.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: THREE PLACE DECIMALS ±.010, ANGLES ±2°.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

APPROVED 24 Apr 51 REVISED (1) 24 Oct 52 (2) 10 Jun 53

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

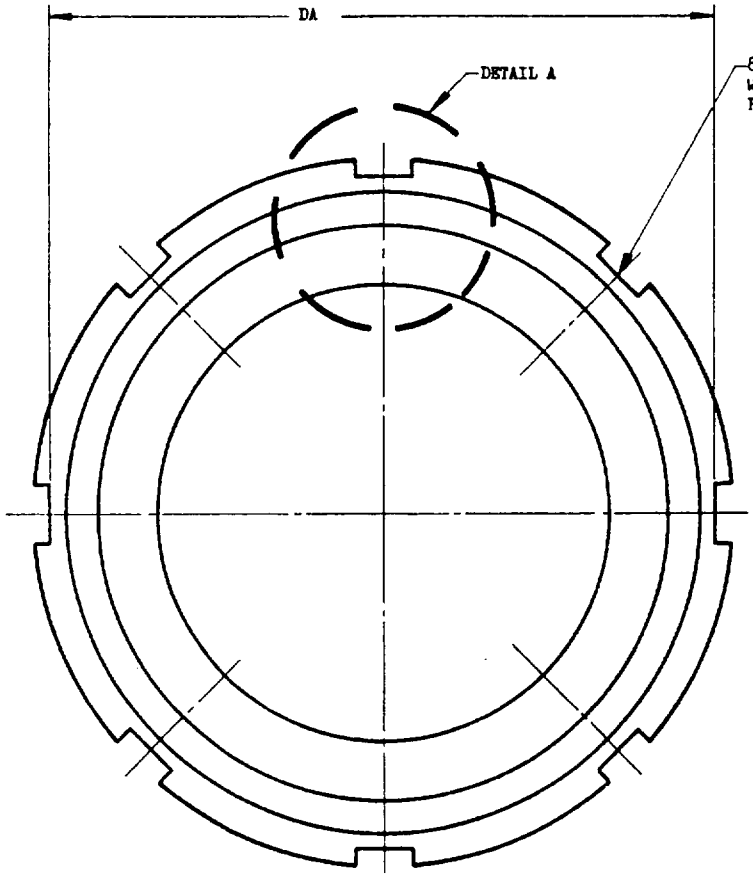
SHAFT ENDS - PROPELLER

AND10152

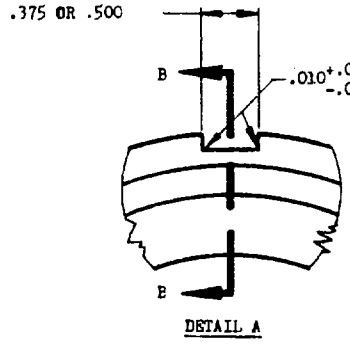
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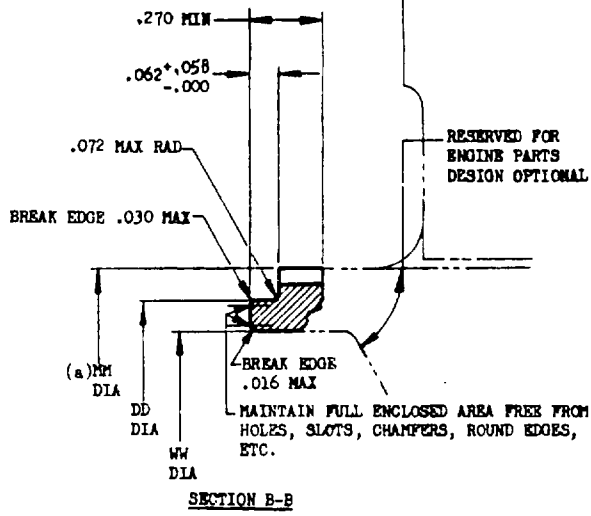
NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all companion standard drawings for the same product and shall become effective for the procurement of commercial supplies, or for use in new designs, not later than 6 months after the latest date of approval shown.



PROPELLER SHAFT THRUST NUT



DETAIL A



SECTION B-B

THRUST NUT

DUAL ROTATION - NO. 41-60A, NO. 51 -70, NO. 60L-80, NO. 65-80, NO. 80-90.

(a) MIN VALUE OF MM APPLIES ONLY TO .270 MIN LENGTH, DETAILED ABOVE, BEYOND WHICH MM MAX ONLY APPLIES AND MM MINIMUM MAY SUIT DESIGN AS FAR AS ENGINE NOSE OUTLINE.

BREAK EDGES .005 - .016 UNLESS OTHERWISE SPECIFIED.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: DECIMALS ±.010, ANGLES ±2°.
(2) FOR OTHER APPLICABLE DIMENSIONS SEE SHEET 4, TABLE II SHEET 13 FOR VALUES OF DIMENSIONS.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

REVISED (2) 10 Jun 53 REVISED (1) 24 Oct 52 REVISED 24 Apr 51

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

SHAFT ENDS – PROPELLER

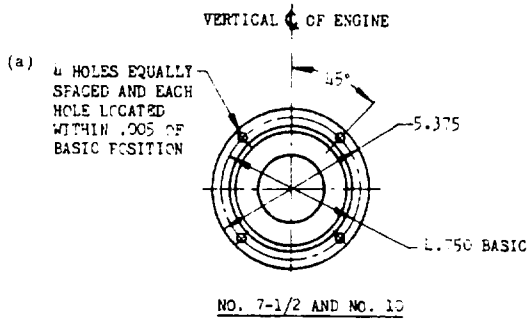
AND10152

SHEET 1 OF 13

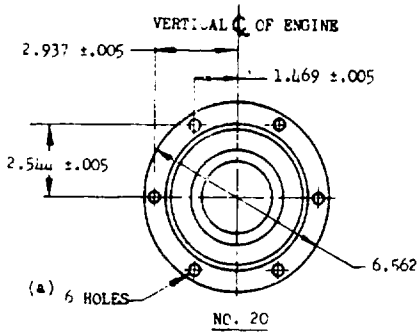
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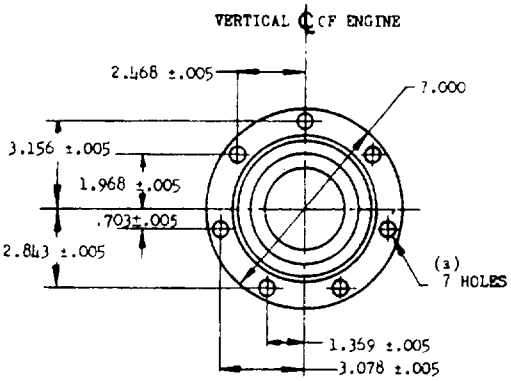
NOTE: This drawing was approved by joint action of the Air Force and Navy Departments on the Air Force-Navy standard for this project. This drawing supersedes all substantial standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new designs, not later than 6 months after the latest dated approval shown.



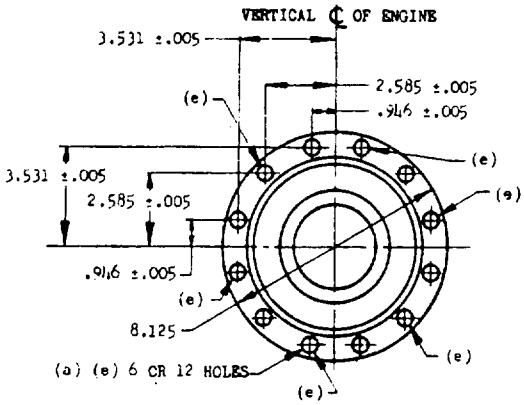
NO. 7-1/2 AND NO. 10



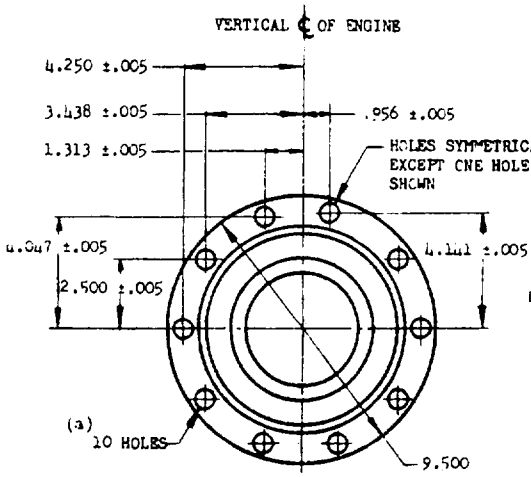
NO. 20



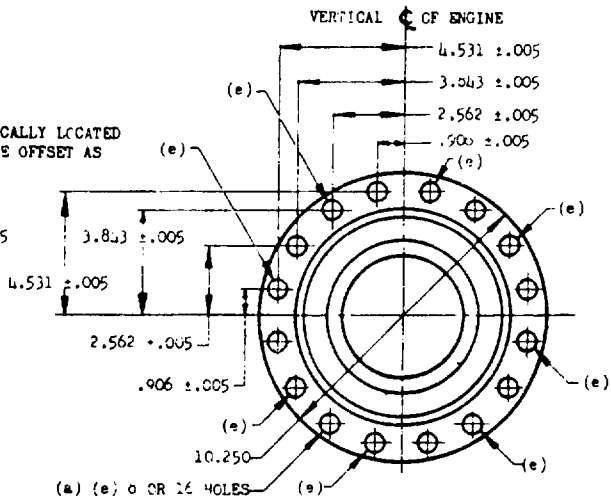
NO. 30



NO. 40 (h) AND NO. 41



NO. 50 (h) AND NO. 51



NO. 60 (h)

PLATE - PROPELLER THRUST BEARING

(a) UNLESS OTHERWISE INDICATED, HOLES ARE SYMMETRICALLY SPACED WITH RESPECT TO HORIZONTAL AND/OR VERTICAL CENTERLINES.

(e) WHEN MINIMUM NUMBER OF HOLES ARE REQUIRED, HOLES MARKED (e) ARE TO BE OMITTED.

(2)(h) INACTIVE FOR DESIGN AFTER 24 APRIL 1951.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

FOR OTHER APPLICABLE DIMENSIONS SEE SHEET 4.

BREAK EDGES .016.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: THREE PLACE DECIMALS ±.010, ANGLES ±2°.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATIONS.

APPROVED 21 Feb 47 REVISED (1) 24 Apr 51 (2) 24 Oct 52

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

SHAFT ENDS - PROPELLER

AND10152

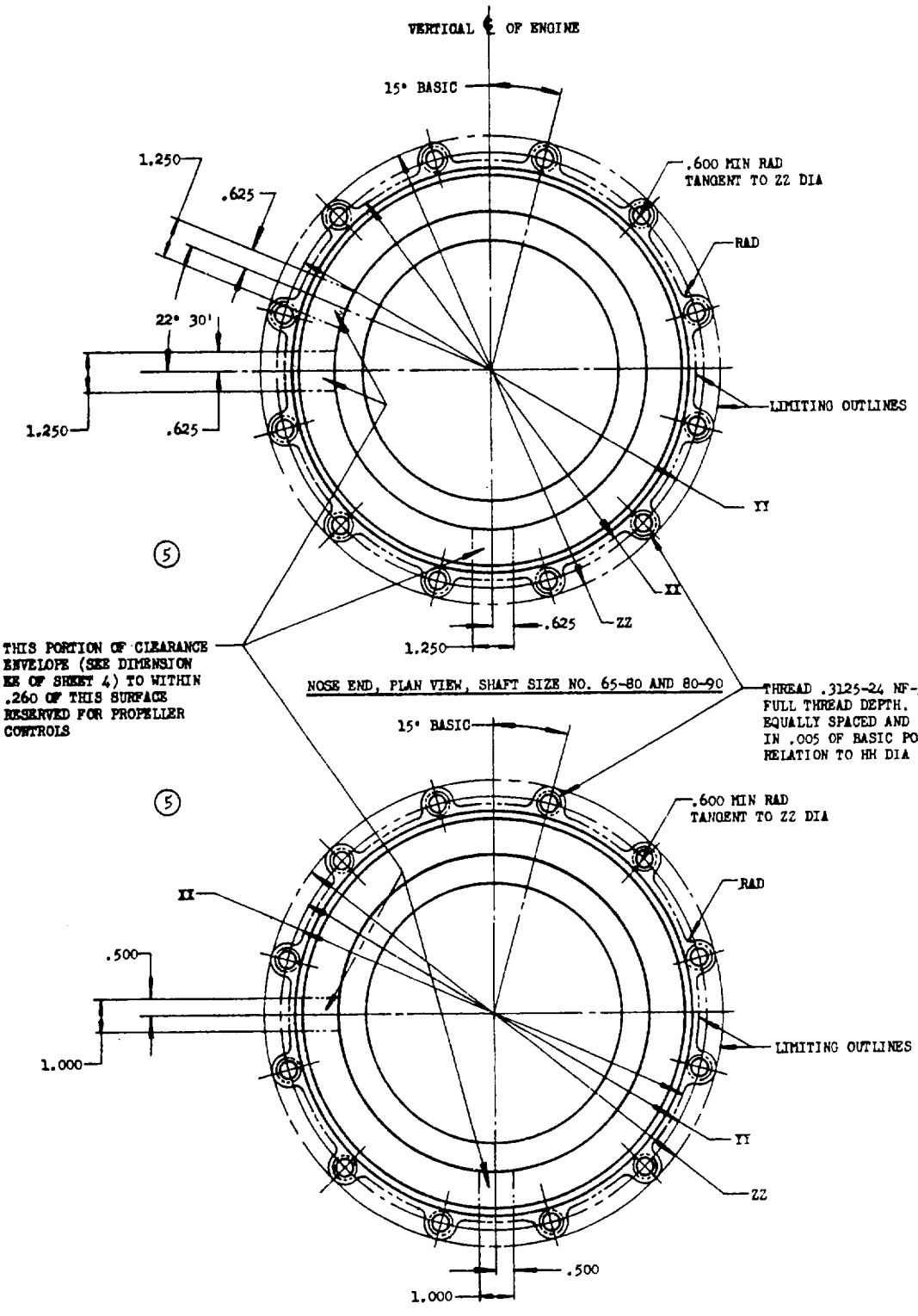
SHEET 7 OF 13

NOT A PART NUMBER

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NOSE ENDS, PLAN VIEW, SHAFT SIZES NO. 60A, 70, 80, 90, 100, 40-60A(h), 41-60A, 50-70(h), 51-70, 60-80(h), 60L-80, 70-90(h)

(h) INACTIVE FOR DESIGN AFTER 24 APRIL 1951.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

BREAK EDGES .016.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: THREE PLACE DECIMALS ±.010, ANGLES ±2°.

SEE TABLES I AND II SHEETS 11 AND 13 FOR VALUES OF DIMENSIONS. FOR OTHER APPLICABLE DIMENSIONS SEE SHEET 4.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATION, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

PROVED 21 Feb 47 REVISED (1) 24 Aug 48 (2) 10 Nov 49 (3) 24 Apr 51 (4) 2 Nov 52 (5) 10 Jun 53

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

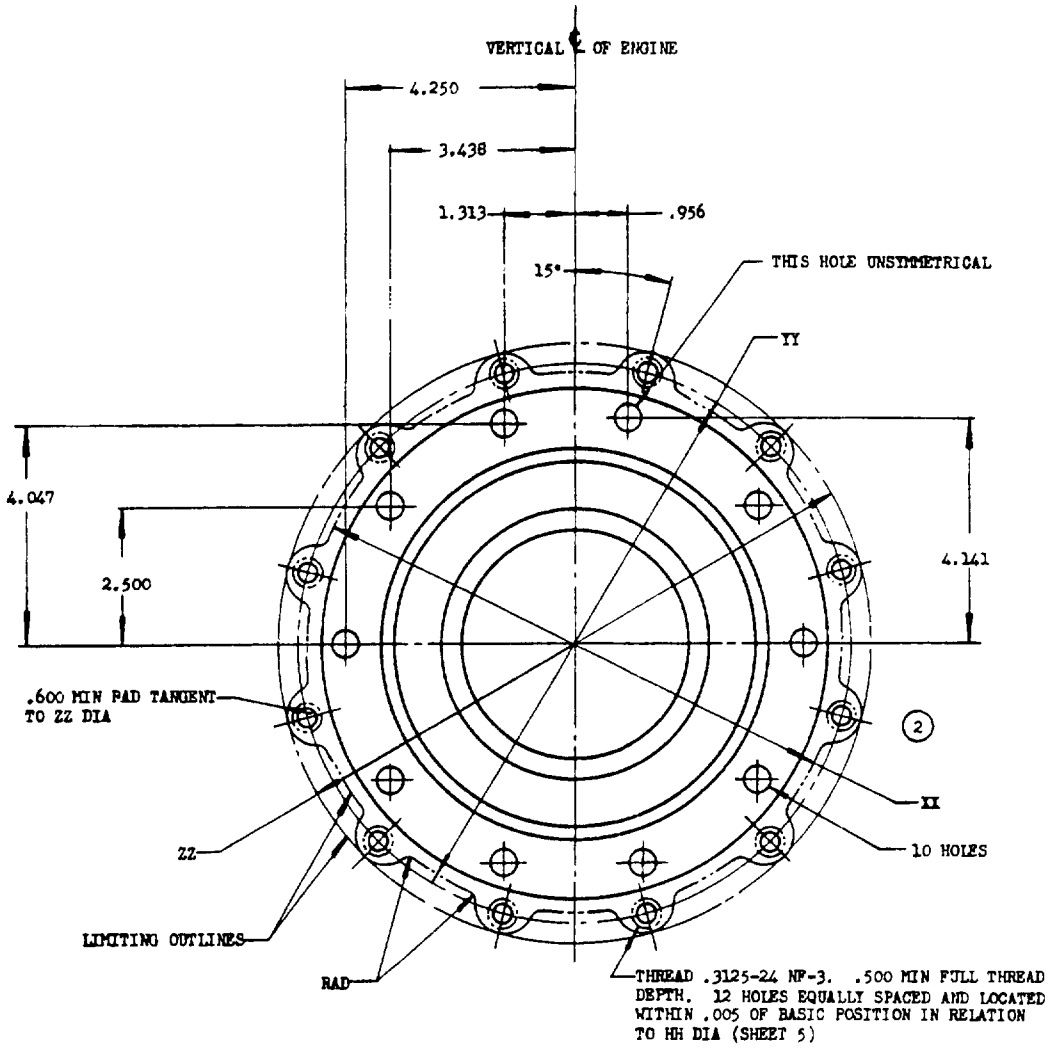
SHAFT ENDS - PROPELLER

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NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all coincident standard drawings for the same product and shall become effective for the procurement of nonassisted supplies, or for use in new designs, not later than 6 months after the latest date of approval shown.



NOSE END, PLAN VIEW, SHAFT SIZES NO. 50A(h) AND NO. 51A

(h) INACTIVE FOR DESIGN AFTER 24 APRIL 1951.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

BREAK EDGES .016.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: THREE PLACE DECIMALS ±.005, ANGLES ±2°.

SEE TABLE 1 SHEET 11 FOR VALUES OF DIMENSIONS. FOR OTHER APPLICABLE DIMENSIONS SEE SHEET 5.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATION, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

APPROVED 24 Apr 51 REVISED (1) 24 Oct 52 (2) 10 Jun 53

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TABLE I SINGLE ROTATION SHAFTS

LETTER	TOLERANCE	(a) 7-1/2	(a) 10	(a) 20	(a) 30	(h) 40 (a)	41	(h) 50 (a)	(h) 50A (a)	51	51A	(h) 60	60A	70	80	90	100
A DIA	+.000 -.002	1.525	1.992	2.367	2.617	3.117	-	3.804		-		-		-	-	-	-
	+.000 -.005	-	-	-	-	-	3.083	-		3.750		4.680		5.539	6.411	7.554	8.667
B DIA	MAX	1.474	1.781	2.156	2.406	2.875	2.917	3.554		3.583		4.436		5.294	6.151	7.294	8.313
C DIA	MIN	1.382	1.689	2.064	2.314	2.783	-	3.462		-		-		-	-	-	-
	+.000 -.020	-	-	-	-	-	2.850	-		3.517		-		-	-	-	-
	+.010 -.020	-	-	-	-	-	-	-		-		4.321		5.179	6.036	7.179	8.199
D	±.0006	.1590	.1940	.2310	.2570	.3040	-	.3750		-		-		-	-	-	-
	+.0000 -.0030	-	-	-	-	-	.1299	-		.1299		.2233		.2233	.2233	.2234	.2607
E DIA	+.000 -.004	1.370	1.682	2.057	2.307	2.807	2.807	3.432		3.432		4.245		5.120	5.995	7.120	8.120
F DIA	+.000 -.005	1.250	1.562	1.938	2.188	2.688	2.688	3.312		3.312		4.062		4.938	5.812	6.938	7.938
G DIA	+.000 -.002	1.375	1.687	2.062	2.312	2.812	2.812	3.500		3.500		4.296		5.156	6.011	7.156	8.174
H DIA	+.000 -.002	1.636	2.000	2.375	2.625	3.125	3.125	3.812		3.812		4.687		5.562	6.426	7.570	8.682
I	±.030	1.375	1.375	2.094	2.087	2.125	2.125	2.062		2.062		3.313		3.312	3.312	3.312	3.500
J	±.040	1.875	1.875	2.625	2.618	2.656	2.656	2.562		2.562		3.812		3.812	3.812	3.812	4.000
K	±.020	4.375	4.688	5.500	5.868	5.469	5.469	5.875		5.875		7.359		8.188	8.938	9.875	10.375
L	±.025	-	-	-	-	5.781	-	6.188		6.030		-		-	-	-	-
M	+.010-(f)	5.402	5.402	6.715	7.115	6.778	6.778	7.152		7.152		8.860		10.360	11.922	13.110	14.735
N	+.010 -.030	6.068	6.068	7.548	7.948	7.611	7.611	8.318		8.318		10.235		11.735	13.297	14.610	16.235
O	±.015	6.375	6.375	7.875	8.243	7.906	7.906	8.562		8.562		10.500		12.125	13.750	15.000	16.625
P	±.020	6.625	6.625	8.125	8.493	8.156	8.156	8.812		8.812		10.750		12.375	14.000	15.250	16.875
Q	±.020	5.312	5.312	6.625	7.025	6.688	6.688	7.062		7.062		8.750		10.250	11.812	13.000	14.625
R	MAX	1.530	1.530	1.530	1.530	1.530	1.530	1.530		1.530		2.030		2.030	2.030	3.030	3.030
	MIN	1.125	1.125	1.125	1.125	1.125	1.125	1.125		1.125		1.125		1.125	1.125	1.125	1.125
S	+.000 -.003	1.319	1.631	2.006	2.256	2.756	2.756	3.381		3.381		-		-	-	-	-
	+.0000 -.0050	-	-	-	-	-	-	-		-		4.1668		5.0418	5.9168	-	-
	+.0000 -.0080	-	-	-	-	-	-	-		-		-		-	-	7.0418	8.0418
T	THREAD	1.3750-12	1.6875-12	2.0625-12	2.3125-12	2.8125-12	2.8125-12	3.4375-12		3.4375-12		4.250-8		5.125-8	6.000-8	7.125-8	8.125-8
U	±.030	.170	.170	.170	.170	.170	.170	.170		.170		.250		.250	.250	.250	.250
V	APPROXIMATE	-	-	-	-	-	.035	-		.035		.068		.068	.068	.068	.080
W	±.0100	.1990	.2656	.2656	.2656	.2656	.2656	.2656		.2656		.2656		.2656	.2656	.2656	.2656
X	-	16	16	16	16	16	36	16		44		32		38	44	52	51
Y		-	-	-	-	-	12/24	-		12/24		7/16		7/16	7/16	7/16	6/12
Z		-	-	-	-	-	3.0000	-		3.6667		4.5714		5.4286	6.2857	7.4286	8.5000

(f) MINUS VALUES DEPENDS ON METHOD OF THREADING AND THREAD RUNOUT RELATION TO SHOULDER Q, M DIMENSION MAX LIMIT GIVES MIN FULL THREAD.

TO OBTAIN DIMENSIONS FOR FULL NUMBER OF PITCHES, WHEN DESIRED, DEDUCT BASIC M FROM BASIC N.

M DIMENSION DOES NOT APPLY WHEN UNDERCUT IS USED.

THIS DRAWING AND APPLICABLE ENGINE SPECIFICATION, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

(h) INACTIVE FOR DESIGN AFTER 24 April 1951.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTE, SEE ANA BULLETIN NO. 337.

③ (a) STRAIGHT SPLINES ONLY.

PROVED 21 Feb 47 REVISED ① 24 Apr 1951 ② 24 Oct 52 ③ 10 Jun 53

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AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

SHAFT ENDS - PROPELLER

AND10152

TABLE I (CONTINUED) SINGLE ROTATION SHAFTS

LETTER	(3) TOLERANCE	7-1/2	10	20	30	(h) 40	41	(h) 50	(h) 50A	51	51A	(h) 60	60A	70	80	90	100
AB DIA	+.020 -.000			1.516	1.766	2.266	2.266	2.891		2.891		3.516					
AC DIA	+.005 -.000			1.435	1.685	2.185	2.185	2.810		2.810		3.413					
AD DIA	+.005 -.000			1.430	1.680	2.180	2.180	2.805		2.805		3.408					
AE DIA	+.020 -.000			1.516	1.766	2.266	2.266	2.891		2.891		3.516					
AF	±.030			.125	.156	.156	.156	.250		.250		.250					
AG	±.010			.500	.562	.562	.562	1.219		1.219		1.219	1.219	1.219	SEE TT	SEE TT	
AH	±.010			.655	.781	.781	.781	1.531		1.531		1.344					
AJ	±.010			.688	.812	.812	.812	1.562		1.562		1.375					
AK	THREAD NS-3			1.500-16	1.750-16	2.250-16	2.250-16	2.875-16		2.875-16		3.500-12					
AL PITCH DIA	+.0040 -.0000			1.4594													
	+.0041 -.0000				1.7094												
	+.0044 -.0000					2.2094	2.2094										
	+.0046 -.0000							2.8344		2.8344							
	+.0053 -.0000											3.4459					
AM	±.015			1.500	1.656	1.656	1.656	2.719		2.719		2.500					
AP	MAX			1.365	1.520	1.520	1.520	2.580		2.580		2.365					
	MIN			1.265	1.420	1.420	1.420	2.480		2.480		2.240					
AQ DIA	MIN			.375	.438	.547	.547	.594		.594		.594					
AR DIA	+.000 -.002			.609	.734	1.047	1.047	1.483		1.483		1.859					
AS DIA	BASIC			1.100	1.350	1.800	1.800	2.400		2.400		2.920					
AT DIA	±.010			.082	.082	.125	.125	.1405		.1405		.1405					
AU	±.010			1.688	1.938	2.438	2.438	3.062		3.062		3.688					
AW RAD	±.010			.375	.375	.500	.500	.500		.500		.500					
AZ	±.010			1.375	1.531	1.531	1.531	2.594		2.594		2.375					
BC DIA	MAX			4.250	4.250	4.750	4.750	6.000		6.000		6.312					
BE	±.010	.188	.188	.312	.312	.375	.375	.375		.375		.375					
BF	±.010								.250		.250						
BH DIA	±.001								7.248		7.248						
BJ	MIN	.115	.115	.250	.250	.125	.125	.344		.344		.219					
BK	NOMINAL	.3125	.3125	.375	.375	.375	.375	.375		.375		.4375	OPTIONAL	OPTIONAL	OPTIONAL	OPTIONAL	OPTIONAL
BL		4	4	6	7	6 OR 12	6 OR 12	10		10		8 OR 16	OPTIONAL	OPTIONAL	OPTIONAL	OPTIONAL	OPTIONAL
BX	MIN			.500	.562	.562	.562	.810		.810		.810					
BZ	MIN	.031	.031	.031	.031	.031	.031	.031		.031		.320					
DD	±.010												5.812	6.688	7.562		
	±.025						4.000		4.875		4.875					8.750	10.000
EE	MAX	.938	.938	.938	.938	1.000	1.000	1.000		1.000		1.016	.750	.750	.750	1.000	1.000
FF	±.010	.188	.188	.188	.188	.188	.188	.250	.312	.250	.312	.250	.312	.312	.312	.312	.375
GG DIA	MIN	3.750	3.750	4.750	4.750	5.875	5.875	6.812		6.812		7.312	7.750	8.750	9.625	11.375	13.000
HH DIA	±.001	3.998	3.998	4.998	4.998	6.248	6.248	7.248	10.123	7.248	10.123	7.748	10.123	11.123	12.625	14.373	15.248
JJ DIA	+.005 -.000			1.625	1.875	2.375	2.375	3.062		3.062		3.625	4.375	5.156	6.250	7.219	
LL	MIN								2.062 (g)		2.062 (g)		2.062 (g)	2.062 (g)	2.062 (g)	4.000	4.000
MM DIA	MAX						4.375		5.250		5.250		6.188	7.062	7.938	9.125	10.875
PP	MIN								35° (g)		35° (g)		35° (g)	35° (g)	35° (g)	45°	45°
QQ	±.035	.312	.312	.344	.025	.172	.172	.406	.718	.406	.718	.625	.938	.938	.938	.938	1.062
SS	MIN								.500		.500		.500	.500	.500	.500	.500
TT	±.030			2.000	2.000	2.000	2.000	3.000		3.000		3.000		3.000	3.000	3.000	3.000
VV DIA	MAX								10.812		10.812		10.102	11.072	12.574	14.250	15.125
XX	MIN								11.250		11.250		10.812	11.812	13.312	15.062	15.938
YY	BASIC								12.250		12.250		11.250	12.250	13.750	15.625	16.375
ZZ	MAX								12.250		12.250		12.250	13.250	14.750	16.750	17.375
	MIN								12.062		12.062		12.062	13.062	14.562	16.562	17.188

(g) LL TO BE 4.000 AND PP TO BE 45° WHEN PRACTICABLE.
(h) INACTIVE FOR DESIGN AFTER 24 APRIL 1951.
FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

AP DIMENSION MAXIMUM LIMIT GIVES MAXIMUM FULL OR IMPERFECT THREAD, MINIMUM LIMIT GIVES MINIMUM FULL THREAD. DIMENSION DOES NOT APPLY WHEN UNDERCUT IS USED (ALL THREADS FULL).

DIMENSIONS IN INCHES.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATION, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

APPROVED 12 Feb 47 REVISED (1) 24 Apr 51 (2) 24 Oct 52 (3) 10 Jun 53

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TABLE II DUAL ROTATION SHAFTS

LETTER	TOLERANCE OUTBOARD SHAFT	TOLERANCE INBOARD SHAFT	(h) 40 — 60A (a)	41 — 60A		(h) 50 — 70 (a)	51 — 70		(h) 60 — 80	60L — 80	65 — 80		(h) 70 — 90	80 — 90	
A DIA	+.000 -.002		3.117	-	-	3.804	-	-	-	-	-	-	-	-	-
	+.000 -.005		-	3.083	4.680	-	3.750	5.539	4.680	4.680	5.250	6.411	5.539	6.411	7.554
B DIA	MAX		2.875	2.917	4.436	3.554	3.583	5.294	4.436	4.436	5.000	6.151	5.294	6.151	7.294
C DIA	MIN		2.783	-	-	3.462	-	-	-	-	-	-	-	-	-
	+.000 -.020		-	2.850	-	-	3.517	-	-	-	4.900	-	-	-	-
	+.010 -.020		-	-	4.321	-	-	5.179	4.321	4.321	-	6.036	5.179	6.036	7.179
D	±.0008		.3040	-	-	.3750	-	-	-	-	-	-	-	-	-
	+.0000 -.0030		-	.1299	.2233	-	.1299	.2233	.2233	.2233	.1953	.2233	.2233	.2233	.2234
E DIA	+.000 -.004		2.807	2.807	4.245	3.432	3.432	5.120	4.245	4.245	4.870	5.995	5.120	5.995	7.120
F DIA	+.000 -.005		2.688	2.688	4.062	3.312	3.312	4.938	4.062	4.062	4.688	5.812	4.938	5.812	6.938
G DIA	+.000 -.002		2.812	2.812	4.296	3.500	3.500	5.156	4.296	4.296	4.875	6.011	5.156	6.011	7.156
H DIA	+.000 -.002		3.125	3.125	4.687	3.812	3.812	5.562	4.687	4.687	5.281	6.426	5.562	6.426	7.570
I	±.020	±.025	4.438	4.438	1.812	7.375	7.375	1.812	8.250	13.250	11.250	3.250	20.188	15.125	3.312
J	±.030	±.040	4.938	4.938	2.312	7.875	7.875	2.312	8.750	13.750	11.750	3.750	20.688	15.625	3.812
K	±.020		8.250	8.250	5.375	11.188	11.188	5.375	12.297	17.297	16.125	7.400	25.062	20.750	8.000
L	±.020	-	8.375	-	-	11.312	11.312	-	-	-	-	-	-	-	-
M	+.000 -(f)	+.010 -(f)	9.475	9.475	8.360	12.412	12.412	8.360	13.808	18.808	18.297	10.360	27.235	23.735	11.610
N	+.010 -.030		10.641	10.641	9.735	13.578	13.578	9.735	15.183	20.183	19.672	11.610	28.610	25.110	13.110
O	±.015	-	10.938	10.938	-	13.812	13.812	-	15.438	20.438	20.062	-	29.000	25.562	-
P	±.020		11.312	11.312	10.625	14.250	14.250	10.625	15.688	20.688	20.312	12.500	29.250	25.812	14.062
Q	±.010	±.020	9.375	9.375	8.250	12.312	12.312	8.250	13.688	18.688	18.188	10.250	27.125	23.625	11.500
R RAD	MAX		1.530	1.530	2.030	1.530	1.530	2.030	2.030	2.030	2.030	2.030	2.030	2.030	3.030
	MIN		1.125	1.125	1.125	1.125	1.125	1.125	1.125	1.125	1.125	1.125	1.125	1.125	1.125
r RAD	+.030 -.000		.062	.062	.062	.062	.062	.062	.062	.062	.062	.062	.062	.062	.062
S PITCH DIA	+.0000 -.0030		2.7560	2.7560	-	3.3810	3.3810	-	-	-	-	-	-	-	-
	+.0000 -.0050		-	-	4.1668	-	-	5.0418	4.1668	4.1668	4.7918	5.9168	5.0418	5.9168	-
	+.0000 -.0080		-	-	-	-	-	-	-	-	-	-	-	-	7.0418
THD T	-		2.9125-12	2.8125-12	4.250-8	3.4375-12	3.4375-12	5.125-8	4.250-8	4.250-8	4.875-8	6.000-8	5.125-8	6.000-8	7.125-8
U	±.030		.170	.170	.250	.170	.170	.250	.250	.250	.250	.250	.250	.250	.250
V RAD	APPROXIMATE		-	.035	.068	-	.035	.068	.068	.068	.060	.068	.068	.068	.068
W DIA	±.0100	-	.2656	.2656	—	.2656	.2656	-	.2656	.2656	.2656	—	.2656	.2656	-
X			16	36	32	16	44	38	32	32	41	44	38	44	52
Y			-	12/24	7/16	-	12/24	7/16	7/16	7/16	8/16	7/16	7/16	7/16	7/16
Z			-	3.0000	4.5714	-	3.6667	5.4286	4.5714	4.5714	5.1250	6.2857	5.4286	6.2857	7.4286

(f) MINUS VALUE DEPENDS ON METHOD OF THREADING AND THREAD RUNOUT RELATION TO SHOULDER Q.

(h) INACTIVE FOR DESIGN AFTER 24 APRIL 1951

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

INBOARD AXIAL DIMENSIONS ARE FROM THRUST NUT.

M DIMENSION MAXIMUM LIMIT GIVES MINIMUM FULL THREAD

TO OBTAIN DIM FOR FULL NUMBER OF PITCHES, WHEN DESIRED, DEDUCT BASIC M FROM BASIC N.

M DIM DOES NOT APPLY WHEN UNDERCUT IS USED.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

(4) (a) STRAIGHT SPLINES ONLY.

APPROVED 21 Feb 47 REVISED (1) 26 Aug 48 (2) 24 Apr 51 (3) 24 Oct 52 (4) 10 Jun 53

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

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SHEET 12 OF 13

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AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

SHAFT ENDS - PROPELLER

AND 0152

SHEET 13 OF 14

NOT A PART NUMBER

TABLE II CONTINUED, DUAL ROTATION SHAFTS

LETTER	TOLERANCE OUTBOARD SHAFT	TOLERANCE INBOARD SHAFT	(h) 40 — 60A (3)			(h) 50 — 70			(h) 60 — 80				(h) 70 — 90		
				41 — 60A			51 — 70			60L — 80	65 — 80			80 — 90	
AA	±.040		10.688	10.688	-	10.688	10.688	-	12.562	12.562	16.062	-	14.125	17.625	-
BB RAD	MAX	-	.094	.094	-	.094	.094	-	.094	.094	.094	-	.094	.094	-
CB	±.020	-	-	-	-	-	-	-	-	-	7.875	-	-	8.875	-
CC	±.020	-	2.375	2.375	-	2.375	2.375	-	4.938	5.938	3.875	-	6.000	4.875	-
CF DIA	+.0000 -.0005	-	-	-	-	-	-	-	-	-	5.3149	-	-	6.4567	-
		-	-	-	-	-	-	-	-	-	(135 MM BRG)	-	-	(164 MM BRG)	-
DA	±.010		-	-	5.880	-	-	6.740	-	-	-	7.600	-	-	8.812
DD DIA	±.010		-	-	5.812	-	-	6.688	-	-	-	7.562	-	-	8.750
EE	MAX		-	-	.750	-	-	.750	-	-	-	.750	-	-	1.000
FF	±.010		-	-	.312	-	-	.312	-	-	-	.312	-	-	.312
GG DIA	MIN		-	-	7.750	-	-	8.750	-	-	-	9.625	-	-	11.375
HH DIA	±.001		-	-	10.123	-	-	11.123	-	-	-	12.625	-	-	14.373
JJ DIA	+.005-.000	-	2.188	2.188	-	2.812	2.812	-	3.562	3.562	4.125	-	4.375	5.156	-
KK DIA	+.0000 -.0005	-	3.1498 (80 MM BRG)	3.1498 (80 MM BRG)	-	3.8191 (97 MM BRG)	3.8191 (97 MM BRG)	-	4.7246 (120 MM BRG)	4.7246 (120 MM BRG)	5.3543 (136 MM BRG)	-	5.5908 (142 MM BRG)	6.4960 (165 MM BRG)	-
LL	(3) MIN		- -	- -	2.062 (k) (3)	- -	- -	(3) 2.062 (k)	- -	- -	- -	(3) 2.062(k) 4.000(m)	- -	- -	4.000
MM DIA	MAX		-	-	6.188	-	-	7.062	-	-	-	7.938	-	-	9.125
	MIN		-	-	6.120	-	-	6.995	-	-	-	7.910	-	-	9.100
NN DIA	MIN	-	3.594	-	-	4.344	-	-	5.375	5.375	-	-	6.250	-	-
	+.000-.010	-	-	3.604	-	-	4.354	-	-	-	5.812	-	-	6.938	-
PP	(3) MIN		- -	- -	(3) 35° (k)	- -	- -	35° (k) (3)	- -	- -	- -	(3) 35° (k) 45° (m)	- -	- -	45°
QQ	±.035		-	-	.938	-	-	.938	-	-	-	.938	-	-	.938
RR DIA	-	±.010		-	3.852	-	-	4.602	-	-	-	5.688	-	-	6.750
SS	MIN		-	-	.500	-	-	.500	-	-	-	.500	-	-	.500
TT	±.030	-	3.000	3.000	-	3.000	3.000	-	3.000	3.000	2.000	-	3.000	2.000	-
UU DIA	-	±.010		-	4.000	-	-	4.812	-	-	-	5.781	-	-	6.875
VV DIA	MAX		-	-	10.102	-	-	11.072	-	-	-	12.574	-	-	14.250
WW DIA	MAX		-	-	5.010	-	-	5.885	-	-	-	6.760	-	-	7.948
XX DIA	MIN		-	-	10.812	-	-	11.812	-	-	-	13.312	-	-	15.062
YY DIA	BASIC		-	-	11.250	-	-	12.250	-	-	-	13.750	-	-	15.625
ZZ DIA	MAX		-	-	12.250	-	-	13.250	-	-	-	14.750	-	-	16.750
	MIN		-	-	12.062	-	-	13.062	-	-	-	14.562	-	-	16.562

(h) INACTIVE FOR DESIGN AFTER 24 April 1951.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

(k) LL TO BE 4.000 AND PP TO BE 45° WHEN PRACTICABLE.

(m) APPLICABLE TO 65-80 SHAFT ONLY.

INBOARD AXIAL DIMENSIONS ARE FROM THE THRUST NUT.

DIMENSIONS IN INCHES.

THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS AND APPLICATION.

APPROVED 21 Feb 47 REVISED (1) 26 Aug 48 (2) 24 Apr 51 (3) 24 Oct 52

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